Linear Algebra | Test for Chapter 4 | November 6, 2026

Name: Richard | R. Hammack | Score: _____

Directions: Put away phones, calculators, smart watches, notes, etc. Show work & answer in the space provided.

1. Write $\mathbf{v} = (10, 1, 4)$ as a linear combination of $\mathbf{u}_1 = (2, 3, 5)$, $\mathbf{u}_2 = (1, 2, 4)$ and $\mathbf{u}_3 = (-2, 2, 3)$.

$(10,1,4) = x(2,3,5) + y(1,2,4) + z(-2,2,3)$

$(10,1,4) = (2x+y-2z,\ 3x+2y+2z,\ 5x+4y+3z)$

$$\begin{cases} 2x+y-2z=10 \\ 3x+2y+2z=1 \\ 5x+4y+3z=4 \end{cases} \qquad \left[\begin{array}{ccc|c} 2&1&-2&10\\3&2&2&1\\5&4&3&4\end{array}\right] \xrightarrow{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 2&1&-2&10\\1&1&4&-9\\5&4&3&4\end{array}\right] \xrightarrow{R_1\leftrightarrow R_2}$$

$$\left[\begin{array}{ccc|c} 1&1&4&-9\\2&1&-2&10\\5&4&3&4\end{array}\right] \xrightarrow[-5R_1+R_3\to R_3]{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1&1&4&-9\\0&-1&-10&28\\0&-1&-17&49\end{array}\right] \xrightarrow[R_2+R_1\to R_1]{-R_2+R_3\to R_3}$$

$$\left[\begin{array}{ccc|c} 1&0&-6&19\\0&-1&-10&28\\0&0&-7&21\end{array}\right] \xrightarrow[-\frac{1}{7}R_3\to R_3]{-R_2\to R_2} \left[\begin{array}{ccc|c} 1&0&-6&19\\0&1&10&-28\\0&0&1&-3\end{array}\right] \xrightarrow{6R_3+R_1\to R_1,\ -10R_3+R_2\to R_2} \left[\begin{array}{ccc|c} 1&0&0&1\\0&1&0&2\\0&0&1&-3\end{array}\right]$$

$$\rightsquigarrow \begin{cases} x=1 \\ y=2 \\ z=-3 \end{cases}$$

Answer: $(10,1,4) = 1(2,3,5) + 2(1,2,4) - 3(-2,2,3)$

$= 1\vec{u}_1 + 2\vec{u}_2 - 3\vec{u}_3$

2. Consider the subset $W = \{f \in P_4 : f(1) = 0\} \subseteq P_4$. Is W a subspace of P_4? Explain.

Note that $f(x) = 0 + 0x + 0x^2 + 0x^3 + 0x^4 \in W$ because $f(1) = 0$.

Now we will check that W is closed under addition. Suppose $f(x), g(x) \in W$, which means $f(1)=0$ and $g(1)=0$. Let $h(x) = f(x) + g(x)$. Notice that $h(1) = f(1) + g(1) = 0 + 0 = 0$, so $h(x) = f(x) + g(x) \in W$. (So W is closed under $+$)

Next we confirm that W is closed under scalar mult. Suppose $f(x) \in W$ and $c \in \mathbb{R}$. Because $f(x) \in W$, we know $f(1) = 0$. Then $cf(1) = c \cdot 0 = 0$, so $cf \in W$, and W is closed under scalar multiplication.

By the above, W is a subspace of P_4

3. Find a basis for the nullspace of the matrix $A = \begin{bmatrix} 1 & 3 & -2 & 4 \\ 0 & 1 & -1 & 2 \\ -2 & -6 & 4 & -8 \end{bmatrix}$.

The nullspace is the set of solutions to $A\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

$$\left[\begin{array}{cccc|c} 1 & 3 & -2 & 4 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ -2 & -6 & 4 & -8 & 0 \end{array}\right] \xrightarrow{2R_1+R_3\to R_3} \left[\begin{array}{cccc|c} 1 & 3 & -2 & 4 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{-3R_2+R_1\to R_1}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & -2 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \rightsquigarrow \begin{cases} w + y - 2z = 0 \\ x - y + 2z = 0 \\ y = (\text{free}) \\ z = (\text{free}) \end{cases} \qquad \begin{cases} w = -y + 2z \\ x = y - 2z \\ y \text{ (free)} \\ z \text{ (free)} \end{cases}$$

Solution set: $\left\{ \begin{bmatrix} -y+2z \\ y-2z \\ y \\ z \end{bmatrix} : y, z \in \mathbb{R} \right\} = \left\{ y\begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + z\begin{bmatrix} 2 \\ -2 \\ 0 \\ 1 \end{bmatrix} : y, z \in \mathbb{R} \right\}$

$= \text{Span}\left(\begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 0 \\ 1 \end{bmatrix} \right)$.

Thus a basis for Null(A) is $\left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 0 \\ 1 \end{bmatrix} \right\}$

4. Suppose the set $S = \{\mathbf{v}, \mathbf{w}\}$ is linearly independent. Without knowing any further information, is it possible to say whether or not the set $S' = \{\mathbf{v}+\mathbf{w},\ \mathbf{v}-\mathbf{w}\}$ is linearly independent or dependent? Explain.

Look at the equation $x(\vec{v}+\vec{w}) + y(\vec{v}-\vec{w}) = \vec{0}$

$$x\vec{v} + x\vec{w} + y\vec{v} - y\vec{w} = \vec{0}$$

$$(x+y)\vec{v} + (x-y)\vec{w} = \vec{0}$$

Because $\vec{v}$ and $\vec{w}$ are linearly independent, the equation $(x+y)\vec{v} + (x-y)\vec{w}$ has only the trivial solution, meaning

$$\begin{cases} x + y = 0 \\ x - y = 0 \end{cases}$$

Adding these equations, $2x = 0 \Rightarrow \boxed{x = 0}$ and back substitution yields $\boxed{y = 0}$.

Conclusion: $x(\vec{v}+\vec{w}) + y(\vec{v}-w) = \vec{0}$ has only the trivial solution, so $S' = \{\vec{v}+\vec{w}, \vec{v}-\vec{w}\}$

5. Suppose a matrix A has 45 columns and its rows are linearly independent.
Also, the set $\{\mathbf{x} \in \mathbb{R}^{45} \mid A\mathbf{x} = \mathbf{0}\}$ is 7-dimensional. How many rows does A have? Explain.

A is an $m \times 45$ matrix, where m is the number of rows.
The rows are linearly independent, so $\boxed{\text{rank}(A) = m}$
Also $N(A) = \{x \in \mathbb{R}^{45} \mid Ax = 0\}$ is 7-dimensional,
so $\boxed{\text{nullity}(A) = 7.}$ The rank-nullity theorem says

$$45 = \text{rank}(A) + \text{nullity}(A)$$
$$45 = \text{rank}(A) + 7$$
$$38 = \text{rank}(A) \leftarrow \text{(\# of rows)}$$

$\boxed{\text{Therefore } A \text{ has } 38 \text{ rows}}$

6. The set $B = \{1 + x + x^2,\ 2 + x + x^2,\ 1 - x + x^2\}$ is a basis for P_2. Consider $3 + x^2 \in P_2$.
Find the coordinate vector $[3 + x^2]_B$.

$$3 + x^2 = c_1(1 + x + x^2) + c_2(2 + x + x^2) + c_3(1 - x + x^2)$$

$$3 + 0x + 1x^2 = (c_1 + 2c_2 + c_3) + (c_1 + c_2 - c_3)x + (c_1 + c_2 + c_3)x^2$$

$$\Rightarrow \begin{cases} c_1 + 2c_2 + c_3 = 3 \\ c_1 + c_2 - c_3 = 0 \\ c_1 + c_2 + c_3 = 1 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 1 & 1 & -1 & 0 \\ 1 & 1 & 1 & 1 \end{array}\right] \begin{array}{l} -R_1 + R_2 \to R_2 \\ -R_1 + R_3 \to R_3 \end{array} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -1 & -2 & -3 \\ 0 & -1 & 0 & -2 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -1 & 0 & -2 \\ 0 & -1 & -2 & -3 \end{array}\right] \begin{array}{l} 2R_2 + R_1 \to R_1 \\ -R_2 + R_3 \to R_3 \end{array} \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & -1 & 0 & -2 \\ 0 & 0 & -2 & -1 \end{array}\right] \begin{array}{l} -R_2 \to R_2 \\ -\frac{1}{2}R_3 \to R_3 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & \frac{1}{2} \end{array}\right] \xrightarrow{-R_3 + R_1 \to R_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -3/2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & \frac{1}{2} \end{array}\right]$$

$c_1 = -3/2$
$c_2 = 2$
$c_3 = 1/2$

Answer: $\boxed{[3 + x^2]_B = \begin{bmatrix} -3/2 \\ 2 \\ 1/2 \end{bmatrix}}$